

Challenge 3 IoT - Exercise

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1 EQ1

Scenario description: A LoRaWAN network in Europe operates at carrier frequency 868 MHz with bandwidth 125 kHz. The network is composed of one gateway and 40 sensor nodes. Each sensor node transmits packets with payload size 20 bytes according to a Poisson process with intensity $\lambda = 2$ packets/minute

Question: Determine the largest LoRa spreading factor (SF) that guarantees a packet success rate of at least 75%

Explanation:

System variables

- Frequency = 868 MHz
- Bandwidth = 125 kHz
- Sensor Nodes = 40
- Payload = 20 Byte
- $\lambda = 2$ packet/minute

To calculate the airtime we used the TTN (The Things Network) airtime calculator using these parameters (EU868, BW = 125 kHz, Input Bytes = 20). Then we calculated the number of packets per second per node like this:

$$\text{Lambda}_s = \lambda / 60 = 0.03333 \quad (1)$$

After that we can calculate the total system rate, which is obtained by multiplying the number of sensors times the number of packets per second per node, in this way:

$$\text{Total_rate} = \text{SensorNodes} \times \text{Lambda}_s = 1.3333 \quad (2)$$

After this, to answer the question we calculate the packet success probability as:

$$P_{\text{success}} = \exp(-2 \times G) \quad (3)$$

where G is the Total_rate times the airtime. We use this formula since an ALOHA pure model describes perfectly a system where packet transmissions are uncoordinated and may overlap over time. Below we have the airtimes, the SF and the packet success rate

SF	T = Airtime (ms)	$G = \lambda \cdot T$	P_success	$\geq 75\%$?
SF7	71.9	0.0959	82.55%	✓ PASS
SF8	133.6	0.1781	70.03%	✗ FAIL
SF9	246.8	0.3291	51.78%	✗ FAIL
SF10	452.6	0.6035	29.91%	✗ FAIL
SF11	987.1	1.3161	7.19%	✗ FAIL
SF12	1810.4	2.4139	0.80%	✗ FAIL

From the table above we can derive that the answer to the first question is SF7 since the largest spreading factor guaranteeing a packet success rate of at least 75% is SF7 ($P_{\text{success}} = 82.55\%$). SF8 already yields $P_{\text{success}} = 70.03\% < 75\%$, so it fails to meet the threshold.

2 EQ2

Scenario description: *After deploying the network, the measured packet success rate is significantly lower than expected. Moreover, performance is not uniform across nodes: some nodes achieve relatively good delivery rates, while others experience very poor performance.*

- *Decrease the number of nodes*
- *Move the nodes closer to the gateway*
- *Increase the spreading factor*

Question: Which of the above actions is the most appropriate? Briefly justify your answer.

Explanation: For us action 2 is the answer, which means moving the nodes closer to the gateway. Nodes far from the gateway suffer higher path loss and may fall below the receiver sensitivity threshold, while nearby nodes function correctly. Moving distant nodes closer restores an adequate SNR margin and equalises performance across the network. Option 1 (fewer nodes) does not address the location-dependent disparity between nodes. Option 3 (higher SF) increases airtime and the number of collisions.